

Supporting Information

The MOBH35 metal-organic barrier heights reconsidered: performance of local-orbital coupled cluster approaches in different static correlation regimes

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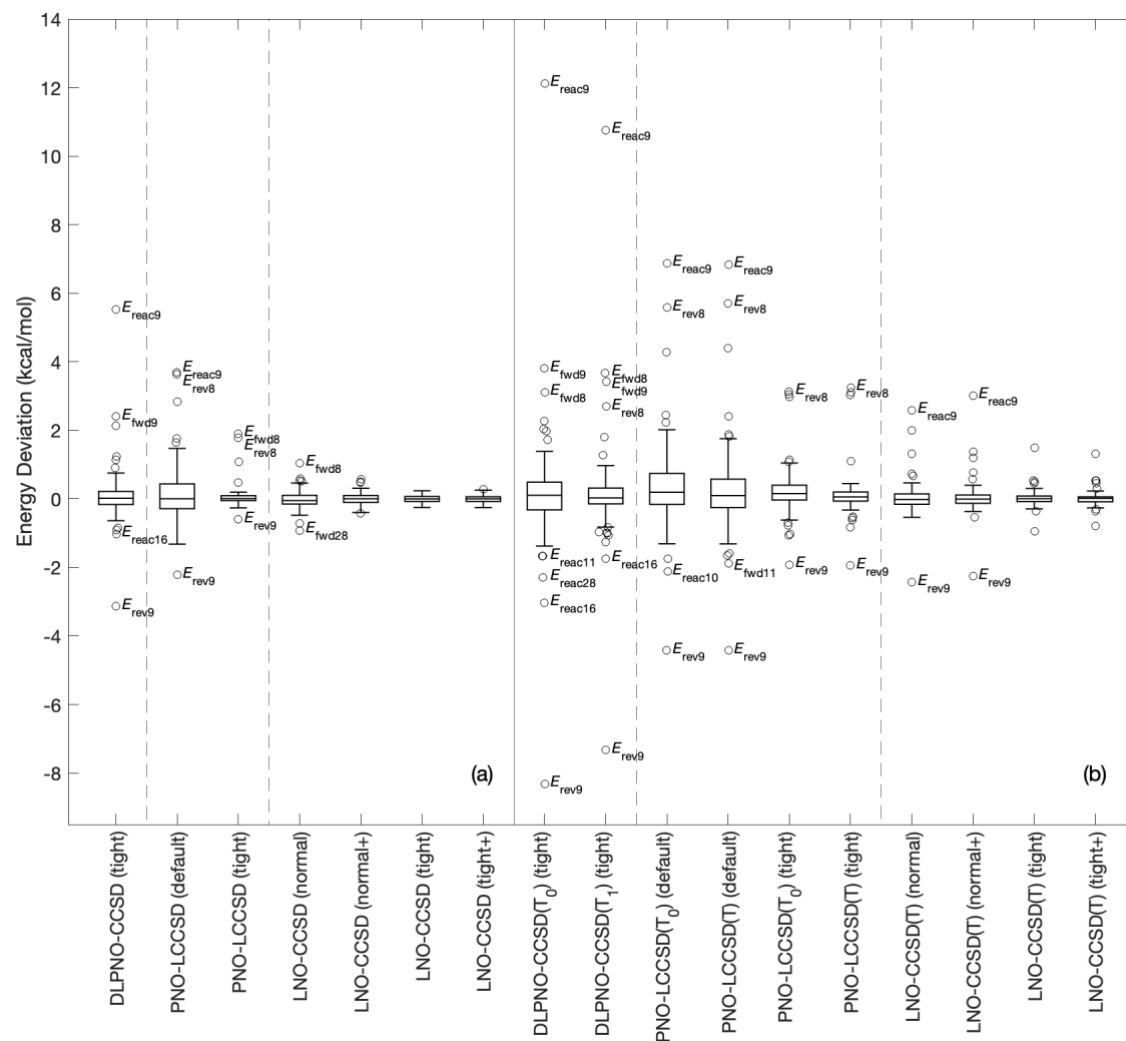


Figure S1. Box-and-whisker plot for the energy deviations of local CC methods (the precision criteria are indicated in parentheses; for LNO-CCSD(T): Normal ($w_{\text{pairtol}} = 1\text{e-}5$), Normal+ ($w_{\text{pairtol}} = 1\text{e-}6$), Tight ($w_{\text{pairtol}} = 3\text{e-}6$), Tight+ ($w_{\text{pairtol}} = 1\text{e-}6$)) from (a) canonical CCSD; (b) canonical CCSD(T). The def2-SV(P) basis set was used throughout.

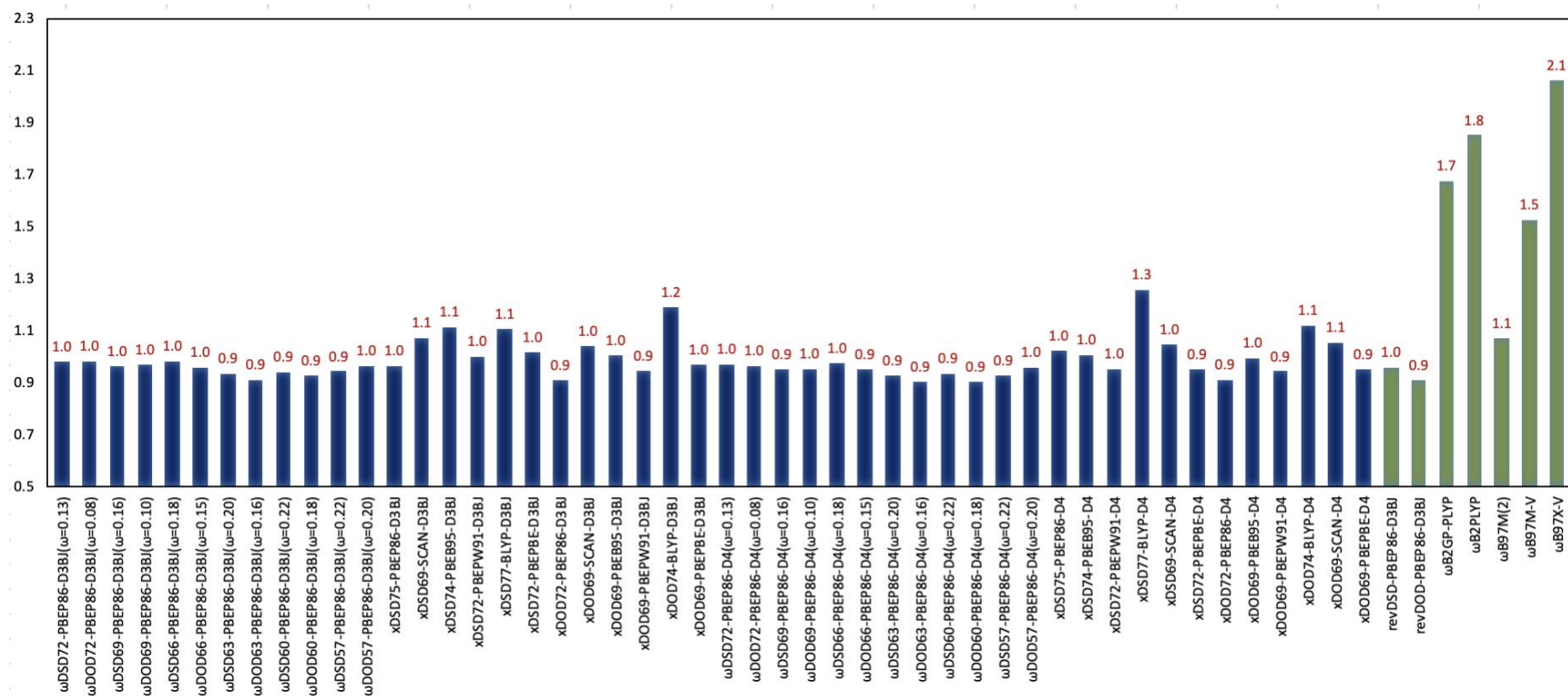


Figure S2. MAD (kcal/mol) statistics for xDSD and ωDSD functionals evaluated against the modified MOBH35 dataset and the revised reference energies reported in this work. Reactions 9, 11, 12, and 17-20 were not included in this comparison.

Table S1. Data for boxplot with quartiles and median from LNO-CCSD(T, Tight+)/def2-TZVPP + $\Delta E_{\text{[CCSD(T)-LNO-CCSD(T)]/def2-SVP}}$ reference values.

			def2-TZVPP	def2-TZVPP	def2-TZVPP	def2-TZVPP	def2-TZVPP	def2-TZVPP	def2-TZVPP	def2-TZVPP	def2-TZVPP	def2-TZVPP
			LNO-CCSD(T) Tight+wpt	LNO-CCSD(T) Tight	LNO-CCSD(T) Normal	LNO-CCSD(T) Normal+	DLPNO-CCSD(T1) Tight T1	DLPNO-CCSD(T) Tight T0	PNO-LCCSD(T) Tight T	PNO-LCCSD(T0) Tight T0	PNO-LCCSD(T) Normal T	PNO-LCCSD(T0) Normal T0
Ref.			LNO Tight+wpt	LNO Tight	LNO Normal	LNO Normal+	DLPNO-T1 Tight	DLPNO-T0 Tight	PNO Tight T	PNO Tight T0	PNO Normal T	PNO Normal T0
26.76	01_end	ΔE#fwd	-0.02	-0.01	0.00	0.05	0.08	0.37	0.02	0.28	-0.38	-0.12
13.94	01_start	ΔE#rev	-0.03	-0.04	-0.05	-0.11	0.20	0.59	-0.16	0.14	-0.54	-0.24
12.82	01_ts	ΔE_R	0.02	0.03	0.05	0.15	-0.12	-0.22	0.18	0.13	0.16	0.12
5.92	02_end	ΔE#fwd	-0.02	0.00	0.14	0.09	-0.06	-0.11	0.13	0.21	0.07	0.15
22.38	02_start	ΔE#rev	-0.10	-0.09	-0.14	-0.11	-0.08	0.14	-0.20	-0.14	-0.46	-0.40
-16.46	02_ts	ΔE_R	0.08	0.09	0.27	0.20	0.03	-0.25	0.33	0.35	0.53	0.55
0.99	03_end	ΔE#fwd	-0.01	-0.01	-0.01	-0.02	-0.01	0.06	0.07	0.15	0.04	0.12
26.29	03_start	ΔE#rev	-0.24	-0.24	-0.52	-0.52	-0.92	-0.02	0.18	0.64	0.12	0.59
-25.31	03_ts	ΔE_R	0.23	0.23	0.51	0.50	0.91	0.08	-0.11	-0.49	-0.07	-0.46
1.53	04_end	ΔE#fwd	0.02	0.02	-0.01	-0.01	0.13	0.00	-0.12	-0.15	-0.16	-0.18
7.81	04_start	ΔE#rev	-0.05	-0.05	-0.17	-0.18	-0.24	0.62	0.19	0.61	0.18	0.60
-6.28	04_ts	ΔE_R	0.07	0.07	0.16	0.17	0.37	-0.62	-0.32	-0.75	-0.34	-0.79
4.75	05_end	ΔE#fwd	-0.06	-0.06	-0.43	-0.29	-0.57	-0.51	-0.18	-0.10	-0.49	-0.42
22.73	05_start	ΔE#rev	-0.09	-0.08	-0.13	-0.14	0.18	-0.18	-0.15	0.03	-0.05	0.12
-17.98	05_ts	ΔE_R	0.03	0.03	-0.30	-0.16	-0.75	-0.33	-0.03	-0.13	-0.44	-0.54
15.60	06_end	ΔE#fwd	-0.14	-0.13	-0.25	-0.20	-0.26	-0.15	0.14	0.38	0.13	0.37
14.76	06_start	ΔE#rev	-0.20	-0.17	-0.37	-0.28	-0.66	-0.72	0.15	0.23	-0.13	-0.05
0.84	06_ts	ΔE_R	0.06	0.04	0.12	0.08	0.40	0.58	-0.01	0.14	0.27	0.42
27.66	07_end	ΔE#fwd	-0.10	-0.08	-0.02	0.05	0.29	0.59	0.17	0.40	0.31	0.53
18.97	07_start	ΔE#rev	-0.10	-0.11	-0.12	-0.06	-0.55	-0.30	-0.18	0.08	0.03	0.27
8.69	07_ts	ΔE_R	0.00	0.04	0.10	0.11	0.84	0.89	0.36	0.33	0.29	0.26
34.69	08_end	ΔE#fwd	0.52		1.60	1.42	3.68	3.01	3.52	3.36	4.54	4.39
31.48	08_start	ΔE#rev	0.19	0.19	0.74	1.18	3.26	2.47	3.67	3.50	6.32	6.13
3.21	08_ts	ΔE_R	0.33		0.86	0.24	0.42	0.54	-0.15	-0.14	-1.78	-1.74
27.83	09_end	ΔE#fwd	0.49	0.49	0.22	0.74	3.70	4.43	1.50	1.54	2.94	2.98
12.67	09_start	ΔE#rev	-1.05	-1.05	-1.98	-1.95	-7.41	-8.23	-1.87	-1.82	-3.82	-3.80

15.16	09_ts	ΔE_R	1.54	1.54	2.19	2.69	11.10	12.66	3.36	3.36	6.76	6.78
-3.79	10_end	ΔE#fwd	-0.18	-0.17	-0.43	-0.44	-0.87	-0.88	-0.52	-0.33	-1.00	-0.80
9.02	10_start	ΔE#rev	0.22	0.26	0.30	0.14	0.33	0.51	0.30	0.78	0.80	1.27
-12.81	10_ts	ΔE_R	-0.40	-0.43	-0.73	-0.58	-1.20	-1.39	-0.82	-1.11	-1.79	-2.08
29.84	11_end	ΔE#fwd	-0.01	0.00	0.02	-0.04	-0.60	-0.53	-0.43	0.01	-0.95	-0.50
82.91	11_start	ΔE#rev	0.00	0.01	0.61	0.67	0.80	1.33	0.44	1.61	0.25	1.38
-53.07	11_ts	ΔE_R	-0.01	-0.01	-0.59	-0.71	-1.40	-1.86	-0.87	-1.60	-1.20	-1.89
5.48	12_end	ΔE#fwd	-0.01	-0.01	-0.22	-0.08	-0.11	-0.13	-0.06	-0.17	-0.06	-0.16
37.23	12_start	ΔE#rev	0.14	0.16	0.27	0.39	-0.07	-0.06	-0.07	-0.20	0.14	0.01
-31.74	12_ts	ΔE_R	-0.15	-0.17	-0.49	-0.47	-0.04	-0.07	0.01	0.03	-0.20	-0.17
20.64	13_end	ΔE#fwd	-0.07	-0.10	-0.02	0.07	1.68	2.19	0.37	0.84	1.33	1.79
48.48	13_start	ΔE#rev	0.14	0.10	0.25	0.33	1.27	2.20	0.49	0.80	1.18	1.48
-27.85	13_ts	ΔE_R	-0.21	-0.19	-0.26	-0.26	0.41	-0.01	-0.12	0.04	0.15	0.31
10.24	14_end	ΔE#fwd	-0.01	-0.01	0.05	0.08	0.21	0.39	0.10	0.26	0.15	0.30
14.44	14_start	ΔE#rev	-0.06	-0.05	-0.03	-0.02	-0.16	-0.22	-0.03	0.15	-0.32	-0.14
-4.21	14_ts	ΔE_R	0.05	0.05	0.08	0.10	0.36	0.61	0.13	0.10	0.46	0.44
20.79	15_end	ΔE#fwd	0.03	0.04	0.04	0.04	-0.07	0.31	0.00	0.26	0.45	0.70
74.68	15_start	ΔE#rev	0.03	0.04	-0.02	-0.04	0.86	2.22	0.75	1.61	1.71	2.56
-53.89	15_ts	ΔE_R	0.00	0.00	0.06	0.08	-0.92	-1.92	-0.75	-1.35	-1.26	-1.86
35.51	16_end	ΔE#fwd	-0.02	-0.02	-0.19	-0.18	-1.40	-2.17	-0.77	-0.67	-1.29	-1.20
53.51	16_start	ΔE#rev	0.08	0.08	0.18	0.12	1.29	1.78	0.32	0.59	0.56	0.83
-18.00	16_ts	ΔE_R	-0.10	-0.10	-0.37	-0.30	-2.70	-3.95	-1.08	-1.26	-1.85	-2.02
8.09	21_end	ΔE#fwd	0.10	0.08	0.32	0.35	0.32	0.47	-0.09	-0.02	-0.33	-0.26
8.10	21_start	ΔE#rev	0.09	0.08	0.32	0.34	0.32	0.48	-0.10	-0.02	-0.34	-0.26
-0.01	21_ts	ΔE_R	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.01	0.01
14.27	22_end	ΔE#fwd	0.18	0.16	0.28	0.32	0.58	0.68	0.43	0.70	0.59	0.85
27.01	22_start	ΔE#rev	0.33	0.31	0.53	0.58	0.94	1.40	0.48	0.87	0.99	1.37
-12.73	22_ts	ΔE_R	-0.15	-0.16	-0.25	-0.26	-0.36	-0.73	-0.05	-0.17	-0.40	-0.52
29.97	23_end	ΔE#fwd	-0.01	N/A	0.26	0.33	0.81	1.36	0.72	1.04	0.63	0.92
20.40	23_start	ΔE#rev	-0.07	N/A	-0.14	-0.04	0.53	1.40	0.47	0.95	1.03	1.47
9.57	23_ts	ΔE_R	0.05	0.06	0.41	0.37	0.28	-0.05	0.25	0.09	-0.40	-0.55

25.42	26_end	ΔE#fwd	-0.09	-0.09	-0.10	-0.11	0.22	0.26	0.01	0.10	0.05	0.14
0.08	26_start	ΔE#rev	0.00	0.01	0.01	0.00	-0.04	-0.01	0.03	0.03	0.01	0.00
25.34	26_ts	ΔE_R	-0.10	-0.09	-0.11	-0.11	0.26	0.27	-0.02	0.07	0.05	0.14
14.03	27_end	ΔE#fwd	0.03	0.04	-0.08	-0.07	0.01	-0.29	0.03	-0.02	0.13	0.08
2.01	27_start	ΔE#rev	-0.08	-0.08	-0.14	-0.11	-0.06	-0.18	-0.13	-0.05	0.09	0.17
12.02	27_ts	ΔE_R	0.12	0.12	0.06	0.05	0.08	-0.11	0.16	0.03	0.04	-0.09
30.47	28_end	ΔE#fwd	0.18	0.18	0.12	0.20	-0.13	-1.25	0.10	0.38	0.00	0.28
15.87	28_start	ΔE#rev	-0.01	-0.01	-0.13	-0.04	0.46	1.17	0.18	0.47	0.25	0.53
14.60	28_ts	ΔE_R	0.19	0.18	0.25	0.24	-0.59	-2.42	-0.08	-0.09	-0.24	-0.25
15.10	29_end	ΔE#fwd	0.16	0.17	0.08	0.17	0.31	0.53	0.28	0.41	0.68	0.80
31.94	29_start	ΔE#rev	0.06	0.03	-0.29	-0.16	0.07	0.07	0.23	0.47	0.56	0.79
-16.84	29_ts	ΔE_R	0.10	0.13	0.37	0.33	0.24	0.46	0.05	-0.06	0.12	0.01
10.04	30_end	ΔE#fwd	0.10	0.11	0.09	0.14	0.02	0.05	0.11	0.28	0.08	0.25
16.92	30_start	ΔE#rev	-0.03	-0.03	-0.03	-0.03	-0.02	0.24	0.06	0.10	-0.35	-0.31
-6.88	30_ts	ΔE_R	0.13	0.14	0.11	0.18	0.04	-0.19	0.05	0.18	0.44	0.56
3.24	31_end	ΔE#fwd	0.07	0.04	-0.24	0.00	0.26	0.13	0.33	0.69	1.49	1.84
13.32	31_start	ΔE#rev	0.05	0.06	0.01	-0.05	0.09	0.56	0.43	0.55	1.25	1.36
-10.08	31_ts	ΔE_R	0.02	-0.02	-0.26	0.05	0.17	-0.43	-0.10	0.14	0.24	0.48
20.58	32_end	ΔE#fwd	-0.08	-0.08	-0.14	-0.12	-0.75	-0.87	-0.07	0.86	-0.19	0.72
61.82	32_start	ΔE#rev	0.18	0.18	0.24	0.20	0.72	0.74	0.38	1.42	0.67	1.69
-41.24	32_ts	ΔE_R	-0.26	-0.26	-0.38	-0.32	-1.48	-1.62	-0.45	-0.56	-0.86	-0.97
1.24	33_end	ΔE#fwd	-0.03	-0.03	-0.11	-0.11	-0.05	0.28	0.04	0.42	-0.02	0.35
8.19	33_start	ΔE#rev	-0.07	-0.07	-0.08	-0.08	-0.24	0.01	0.02	0.52	0.00	0.49
-6.95	33_ts	ΔE_R	0.04	0.04	-0.03	-0.03	0.20	0.27	0.03	-0.10	-0.01	-0.14
29.76	34_end	ΔE#fwd	-0.03	-0.02	0.01	-0.07	0.02	0.51	0.46	0.55	0.96	1.05
3.42	34_start	ΔE#rev	0.01	0.01	0.02	0.06	0.02	0.29	0.23	0.41	0.49	0.67
26.33	34_ts	ΔE_R	-0.04	-0.03	-0.01	-0.13	0.00	0.23	0.22	0.14	0.47	0.38
16.74	35_end	ΔE#fwd	0.00	0.00	0.20	0.23	0.63	0.91	0.17	0.42	0.53	0.77
-2.58	35_start	ΔE#rev	0.08	0.10	0.19	0.07	0.34	0.42	0.30	0.18	0.87	0.74
19.32	35_ts	ΔE_R	-0.08	-0.10	0.01	0.17	0.29	0.49	-0.13	0.24	-0.34	0.02

Table S1. Data for boxplot with quartiles and median from LNO-CCSD(T, Tight+)/def2-TZVPP + $\Delta E_{[\text{CCSD(T)}-\text{LNO-CCSD(T)}]/\text{def2-SVP}}$ reference values (continued)

	MAX	1.54	1.54	2.19	2.69	11.10	12.66	3.67	3.50	6.76	6.78
	97.5%	0.47	0.31	0.84	1.11	3.62	2.93	3.08	3.10	4.30	4.18
	%90	0.19	0.18	0.34	0.36	0.88	1.40	0.46	0.90	1.09	1.47
box top	75%	0.08	0.08	0.18	0.17	0.37	0.57	0.27	0.50	0.51	0.73
median	50%	0.00	0.00	0.01	0.00	0.08	0.13	0.05	0.15	0.08	0.25
x	AVG	0.02	0.02	0.03	0.06	0.19	0.22	0.15	0.28	0.24	0.36
box bottom	25%	-0.07	-0.07	-0.14	-0.11	-0.12	-0.22	-0.10	-0.04	-0.33	-0.21
	10%	-0.14	-0.15	-0.37	-0.29	-0.80	-1.03	-0.36	-0.39	-0.90	-0.79
	2.5%	-0.26	-0.26	-0.58	-0.57	-1.47	-2.38	-0.86	-1.34	-1.79	-2.00
	MIN	-1.05	-1.05	-1.98	-1.95	-7.41	-8.23	-1.87	-1.82	-3.82	-3.80
RMSD		0.25	0.24	0.45	0.48	1.70	1.98	0.77	0.88	1.39	1.47
IQR (interquartile range)		0.15	0.16	0.33	0.29	0.49	0.79	0.36	0.53	0.84	0.94
top whisker		0.31	0.31	0.67	0.60	1.10	1.75	0.81	1.29	1.76	2.14
bottom whisker		-0.29	-0.31	-0.63	-0.54	-0.86	-1.40	-0.64	-0.83	-1.58	-1.63
top extreme fence		0.53	0.55	1.16	1.03	1.84	2.93	1.36	2.09	3.02	3.56
bottom extreme fence		-0.52	-0.54	-1.12	-0.97	-1.60	-2.59	-1.19	-1.63	-2.83	-3.04
		LNO Tight+wpt	LNO Tight	LNO Normal	LNO Normal+	DLPNO-T1 Tight	DLPNO-T0 Tight	PNO Tight T	PNO Tight T0	PNO Normal T	PNO Normal T0

Table S2. Data for boxplot with quartiles and median from CCSD(T)/def2-SVP reference values.

			def2-SVP LNO- CCSD(T)	def2-SVP LNO- CCSD(T)	def2-SVP LNO- CCSD(T)	def2-SVP LNO- CCSD(T)	def2-SVP DLPNO- CCSD(T1)	def2-SVP DLPNO- CCSD(T)	def2-SVP DLPNO- CCSD(T1)	def2-SVP PNO-LCCSD(T)	def2-SVP PNO- LCCSD(T0)	def2-SVP PNO- LCCSD(T)	def2-SVP PNO- LCCSD(T0)
			VeryTight LNO VeryTight	Tight+wpt LNO Tight+wpt	Tight LNO Tight	Normal LNO Normal	Tight T1 DLPNO-T1 Tight	Tight T0 DLPNO-T0 Tight	Normal T1 DLPNO-T1 Normal	Tight T1 PNO Tight	Tight T0 PNO-T0 Tight	Normal T1 PNO Normal	Normal T0 PNO-T0 Normal
Ref.													
27.06	01_end	ΔE_{fwd}	-0.02	-0.01	-0.01	-0.11	0.12	0.39	0.03	0.12	0.36	-0.19	0.05
14.02	01_start	ΔE_{rev}	0.01	-0.02	-0.03	-0.16	0.24	0.62	0.62	0.05	0.33	-0.55	-0.26
13.04	01_ts	ΔE_R	-0.04	0.02	0.02	0.05	-0.12	-0.23	-0.58	0.07	0.02	0.36	0.32
5.63	02_end	ΔE_{fwd}	-0.01	-0.03	-0.02	0.06	-0.05	-0.09	0.13	0.09	0.17	0.04	0.12
25.10	02_start	ΔE_{rev}	0.00	-0.11	-0.10	-0.14	0.05	0.25	0.11	0.04	0.12	-0.38	-0.30
-19.47	02_ts	ΔE_R	-0.01	0.08	0.08	0.20	-0.09	-0.34	0.01	0.05	0.05	0.42	0.42
0.95	03_end	ΔE_{fwd}	0.01	0.00	0.00	0.02	0.01	0.08	0.06	0.08	0.15	0.08	0.15
27.07	03_start	ΔE_{rev}	-0.11	-0.23	-0.24	-0.43	-0.55	0.38	-1.44	0.34	0.86	0.35	0.88
-26.12	03_ts	ΔE_R	0.12	0.23	0.23	0.45	0.56	-0.30	1.50	-0.26	-0.71	-0.27	-0.73
2.36	04_end	ΔE_{fwd}	0.02	0.02	0.02	-0.01	-0.07	-0.20	0.23	-0.15	-0.17	-0.14	-0.17
8.60	04_start	ΔE_{rev}	-0.06	-0.05	-0.05	0.01	-0.24	0.70	-0.59	0.13	0.62	0.19	0.68
-6.24	04_ts	ΔE_R	0.08	0.06	0.06	-0.02	0.17	-0.90	0.82	-0.28	-0.79	-0.33	-0.84
4.68	05_end	ΔE_{fwd}	-0.04	-0.04	-0.06	-0.29	-0.67	-0.68	-0.92	0.00	0.03	-0.56	-0.52
22.02	05_start	ΔE_{rev}	-0.03	-0.06	-0.07	-0.16	0.14	-0.21	0.22	0.18	0.30	0.31	0.43
-17.35	05_ts	ΔE_R	-0.02	0.02	0.01	-0.14	-0.81	-0.47	-1.14	-0.18	-0.27	-0.87	-0.95
13.44	06_end	ΔE_{fwd}	-0.09	-0.12	-0.14	0.05	0.10	0.22	0.59	0.18	0.37	0.30	0.48
13.56	06_start	ΔE_{rev}	-0.12	-0.21	-0.19	-0.28	-0.62	-0.74	-0.59	-0.02	0.02	0.05	0.08
-0.12	06_ts	ΔE_R	0.04	0.09	0.05	0.33	0.72	0.97	1.19	0.20	0.36	0.25	0.41
26.66	07_end	ΔE_{fwd}	-0.10	-0.11	-0.09	-0.08	0.09	0.30	-0.14	0.08	0.30	0.92	1.11
18.30	07_start	ΔE_{rev}	-0.10	-0.10	-0.11	-0.25	-0.88	-0.73	-0.48	-0.09	0.13	0.42	0.62
8.36	07_ts	ΔE_R	0.00	-0.01	0.02	0.17	0.97	1.03	0.34	0.17	0.17	0.50	0.49
36.92	08_end	ΔE_{fwd}	0.26	0.52	0.53	2.01	3.94	3.28	4.17	3.23	3.10	4.63	4.50
32.30	08_start	ΔE_{rev}	-0.05	0.18	0.19	1.09	2.97	2.19	3.51	3.27	3.14	5.74	5.59
4.62	08_ts	ΔE_R	0.31	0.34	0.34	0.92	0.97	1.09	0.66	-0.04	-0.04	-1.11	-1.09
28.59	09_end	ΔE_{fwd}	0.24	0.50	0.49	0.19	3.59	4.09	4.67	1.21	1.26	2.80	2.84

15.20	09_start	ΔE#rev	-0.33	-0.94	-1.06	-2.23	-7.50	-8.55	-10.34	-2.01	-1.99	-4.30	-4.30
13.39	09_ts	ΔE_R	0.57	1.44	1.55	2.42	11.09	12.63	15.02	3.23	3.25	7.10	7.14
-3.48	10_end	ΔE#fwd	-0.14	-0.20	-0.18	-0.06	-0.83	-0.87	-1.01	-0.55	-0.43	-1.08	-0.95
9.58	10_start	ΔE#rev	0.17	0.19	0.22	0.20	0.18	0.36	1.05	-0.18	0.43	0.37	0.97
-13.06	10_ts	ΔE_R	-0.31	-0.39	-0.40	-0.26	-1.01	-1.23	-2.06	-0.37	-0.86	-1.45	-1.91
29.81	11_end	ΔE#fwd	-0.10	-0.02	-0.01	0.04	-0.38	-0.28	-0.23	-0.62	-0.03	-1.68	-1.10
84.09	11_start	ΔE#rev	-0.05	-0.02	0.00	0.39	0.60	1.13	2.69	0.18	1.03	-0.28	0.53
-54.27	11_ts.	ΔE_R	-0.05	0.00	-0.01	-0.35	-0.98	-1.41	-2.92	-0.81	-1.07	-1.41	-1.64
5.67	12_end	ΔE#fwd	0.01	0.00	-0.01	-0.18	-0.18	-0.20	-0.30	-0.05	-0.16	0.09	-0.02
36.83	12_start	ΔE#rev	-0.04	0.16	0.14	-0.04	-0.18	-0.23	-1.01	-0.09	-0.19	0.09	-0.01
-31.16	12_ts	ΔE_R	0.05	-0.16	-0.15	-0.14	0.00	0.03	0.70	0.04	0.04	0.01	0.00
18.37	13_end	ΔE#fwd	0.00	-0.08	-0.07	-0.01	1.80	2.34	2.69	0.39	0.85	1.83	2.26
48.18	13_start	ΔE#rev	0.07	0.13	0.14	0.11	1.29	2.08	1.34	0.47	0.78	1.15	1.44
-29.81	13_ts	ΔE_R	-0.06	-0.21	-0.21	-0.12	0.51	0.25	1.35	-0.08	0.07	0.68	0.82
10.17	14_end	ΔE#fwd	-0.07	0.00	-0.01	0.00	0.15	0.32	0.37	0.12	0.23	0.28	0.38
13.38	14_start	ΔE#rev	-0.06	-0.06	-0.06	-0.13	-0.16	-0.21	0.23	-0.03	0.12	-0.22	-0.08
-3.21	14_ts	ΔE_R	-0.01	0.05	0.05	0.12	0.30	0.53	0.14	0.15	0.11	0.50	0.46
23.90	15_end	ΔE#fwd	0.00	0.03	0.03	0.06	0.11	0.43	0.25	0.10	0.37	0.68	0.94
74.84	15_start	ΔE#rev	0.00	0.03	0.03	-0.02	0.45	1.85	1.55	0.33	1.09	1.29	2.02
-50.94	15_ts	ΔE_R	0.00	0.00	0.00	0.08	-0.34	-1.42	-1.30	-0.23	-0.72	-0.60	-1.08
37.45	16_end	ΔE#fwd	-0.02	-0.01	-0.02	0.01	-0.76	-1.71	-1.96	-0.34	-0.33	-0.87	-0.85
55.56	16_start	ΔE#rev	0.01	0.07	0.08	0.15	1.03	1.53	1.88	0.12	0.38	0.20	0.45
-18.11	16_ts	ΔE_R	-0.03	-0.08	-0.10	-0.14	-1.79	-3.24	-3.83	-0.46	-0.70	-1.08	-1.30
11.11	21_end	ΔE#fwd	0.01	0.10	0.10	0.19	0.48	0.67	0.56	-0.06	0.09	-0.34	-0.20
11.11	21_start	ΔE#rev	0.00	0.09	0.09	0.19	0.44	0.63	0.46	-0.06	0.09	-0.34	-0.20
0.00	21_ts	ΔE_R	0.00	0.00	0.00	0.00	0.04	0.04	0.10	0.00	0.00	0.00	0.00
14.86	22_end	ΔE#fwd	0.00	0.20	0.18	0.31	0.04	0.15	0.34	0.23	0.48	0.96	1.18
30.88	22_start	ΔE#rev	0.13	0.35	0.33	0.47	0.69	1.14	0.75	0.36	0.65	1.51	1.78
-16.02	22_ts	ΔE_R	-0.13	-0.15	-0.15	-0.16	-0.64	-0.99	-0.41	-0.12	-0.17	-0.55	-0.60
29.48	23_end	ΔE#fwd	-0.07	-0.01	-0.01	0.17	0.70	1.14	2.62	0.73	1.01	0.78	1.03
20.80	23_start	ΔE#rev	-0.06	-0.06	-0.07	-0.22	0.43	1.24	0.98	0.29	0.74	0.92	1.35

8.68	23_ts	ΔE_R	-0.02	0.05	0.05	0.40	0.26	-0.10	1.64	0.45	0.27	-0.15	-0.31
21.92	26_end	ΔE#fwd	0.02	-0.11	-0.09	0.04	0.28	0.38	0.60	0.07	0.14	0.12	0.19
-0.07	26_start	ΔE#rev	-0.01	0.00	0.00	0.07	0.00	-0.01	-0.03	0.03	0.03	-0.03	-0.03
21.99	26_ts	ΔE_R	0.03	-0.10	-0.10	-0.03	0.28	0.38	0.64	0.03	0.11	0.15	0.22
16.09	27_end	ΔE#fwd	0.01	0.04	0.03	0.06	0.06	-0.18	0.09	0.11	0.09	0.17	0.15
1.29	27_start	ΔE#rev	-0.05	-0.07	-0.08	-0.08	0.05	0.00	-0.32	-0.05	0.00	0.31	0.37
14.81	27_ts	ΔE_R	0.06	0.11	0.12	0.14	0.01	-0.18	0.40	0.17	0.08	-0.14	-0.22
31.96	28_end	ΔE#fwd	0.06	0.18	0.18	0.21	-0.32	-0.98	-1.29	0.25	0.48	0.11	0.34
16.85	28_start	ΔE#rev	-0.02	-0.01	-0.01	-0.02	0.51	1.08	0.95	0.19	0.40	0.23	0.43
15.11	28_ts	ΔE_R	0.08	0.19	0.19	0.23	-0.83	-2.06	-2.24	0.06	0.08	-0.12	-0.09
15.74	29_end	ΔE#fwd	0.04	0.18	0.16	-0.11	0.21	0.38	0.42	0.24	0.31	0.51	0.57
33.86	29_start	ΔE#rev	0.02	0.10	0.06	-0.10	-0.13	-0.16	0.01	0.33	0.48	0.49	0.64
-18.12	29_ts	ΔE_R	0.02	0.07	0.10	-0.01	0.34	0.54	0.41	-0.09	-0.17	0.02	-0.06
10.87	30_end	ΔE#fwd	0.03	0.09	0.10	-0.03	-0.01	0.01	-0.07	0.04	0.16	0.06	0.18
19.87	30_start	ΔE#rev	-0.01	0.00	-0.01	-0.05	-0.12	0.10	-0.03	-0.01	0.04	-0.64	-0.58
-8.99	30_ts	ΔE_R	0.04	0.09	0.11	0.01	0.10	-0.10	-0.05	0.05	0.12	0.69	0.76
2.14	31_end	ΔE#fwd	0.03	0.09	0.07	-0.38	0.32	0.34	-0.36	0.28	0.49	2.36	2.56
12.37	31_start	ΔE#rev	0.00	0.05	0.05	-0.06	-0.09	0.20	0.98	0.20	0.22	2.18	2.20
-10.23	31_ts	ΔE_R	0.04	0.05	0.02	-0.32	0.41	0.14	-1.34	0.07	0.27	0.17	0.36
23.66	32_end	ΔE#fwd	-0.05	-0.08	-0.08	-0.09	-0.56	-0.83	-0.88	0.09	0.57	-0.02	0.44
58.44	32_start	ΔE#rev	0.04	0.18	0.18	0.10	0.35	0.34	0.72	0.01	0.54	0.09	0.61
-34.78	32_ts	ΔE_R	-0.09	-0.26	-0.26	-0.19	-0.91	-1.17	-1.61	0.08	0.03	-0.12	-0.17
2.77	33_end	ΔE#fwd	-0.02	-0.03	-0.03	-0.03	0.03	0.19	0.27	0.06	0.33	0.04	0.29
9.96	33_start	ΔE#rev	-0.03	-0.07	-0.07	-0.03	-0.07	0.19	-0.13	0.07	0.46	0.05	0.42
-7.20	33_ts	ΔE_R	0.01	0.04	0.04	0.00	0.10	0.00	0.40	-0.01	-0.13	-0.01	-0.13
28.85	34_end	ΔE#fwd	-0.03	-0.05	-0.03	-0.02	-0.21	0.07	0.67	0.20	0.23	1.08	1.11
4.32	34_start	ΔE#rev	-0.01	0.02	0.01	0.01	-0.05	0.23	-0.49	0.19	0.33	0.52	0.65
24.53	34_ts	ΔE_R	-0.02	-0.06	-0.04	-0.03	-0.16	-0.16	1.16	0.01	-0.10	0.56	0.45
14.97	35_end	ΔE#fwd	0.00	-0.01	0.00	0.17	0.63	1.02	2.02	0.16	0.38	0.40	0.61
-3.85	35_start	ΔE#rev	0.06	0.05	0.08	0.06	0.24	0.31	0.92	0.18	0.12	1.09	1.03
18.82	35_ts	ΔE_R	-0.06	-0.06	-0.08	0.11	0.39	0.72	1.10	-0.02	0.26	-0.69	-0.43

Table S2. Data for boxplot with quartiles and median from CCSD(T)/def2-SVP reference values (continued)

	MAX	0.57	1.44	1.55	2.42	11.09	12.63	15.02	3.27	3.25	7.10	7.14
	97.5%	0.26	0.47	0.47	1.07	3.50	3.14	4.07	2.92	2.83	4.35	4.25
	%90	0.07	0.18	0.19	0.32	0.82	1.18	1.73	0.35	0.81	1.21	1.58
box top	75%	0.02	0.09	0.08	0.13	0.40	0.58	0.87	0.19	0.39	0.50	0.66
median	50%	0.00	0.00	0.00	0.00	0.06	0.15	0.23	0.07	0.15	0.11	0.34
X	AVG	0.00	0.02	0.02	0.05	0.20	0.22	0.31	0.15	0.25	0.31	0.41
box bottom	25%	-0.05	-0.06	-0.07	-0.11	-0.17	-0.22	-0.38	-0.05	0.02	-0.20	-0.18
	10%	-0.09	-0.13	-0.14	-0.23	-0.78	-0.98	-1.32	-0.24	-0.29	-0.76	-0.89
	2.5%	-0.14	-0.26	-0.26	-0.37	-1.00	-2.01	-2.82	-0.61	-0.85	-1.45	-1.59
	MIN	-0.33	-0.94	-1.06	-2.23	-7.50	-8.55	-10.34	-2.01	-1.99	-4.30	-4.30
RMSD		0.11	0.23	0.25	0.47	1.67	1.94	2.38	0.69	0.77	1.42	1.48
IQR (interquartile range)		0.07	0.15	0.15	0.24	0.57	0.80	1.26	0.23	0.37	0.71	0.85
top whisker		0.13	0.31	0.31	0.50	1.26	1.78	2.76	0.54	0.94	1.57	1.94
bottom whisker		-0.16	-0.28	-0.29	-0.47	-1.03	-1.42	-2.27	-0.40	-0.53	-1.26	-1.45
top extreme fence		0.24	0.53	0.53	0.86	2.11	2.98	4.64	0.89	1.49	2.63	3.21
bottom extreme fence		-0.26	-0.50	-0.52	-0.84	-1.89	-2.61	-4.15	-0.75	-1.08	-2.33	-2.73
	LNO VeryTight	LNO Tight+wpt	LNO Tight	LNO Normal	DLPNO-T1 Tight	DLPNO-T0 Tight	DLPNO-T1 Normal	PNO Tight	PNO-T0 Tight	PNO Normal	PNO-T0 Normal	

Table S3. Frozen core electrons per molecule in MOBH35 for all post Hartree-Fock calculations.

Molecule	Number of frozen core electrons
P ₁	64
R ₁	64
TS ₁	64
P ₂	56
R ₂	56
TS ₂	56
P ₃	30
R ₃	30
TS ₃	30
P ₄	28
R ₄	28
TS ₄	28
P ₅	72
R ₅	72
TS ₅	72
P ₆	58
R ₆	58
TS ₆	58
P ₇	58
R ₇	58
TS ₇	58
P ₈	76
R ₈	76
TS ₈	76
P ₉	76
R ₉	76
TS ₉	76
P ₁₀	40
R ₁₀	44
TS ₁₀	44
P ₁₁	56
R ₁₁	56
TS ₁₁	56
P ₁₂	66
R ₁₂	66
TS ₁₂	66

P ₁₃	64
R ₁₃	64
TS ₁₃	64
P ₁₄	42
R ₁₄	42
TS ₁₄	42
P ₁₅	30
R ₁₅	30
TS ₁₅	30
P ₁₆	20
R ₁₆	20
TS ₁₆	20
P ₂₁	30
R ₂₁	30
TS ₂₁	30
P ₂₂	42
R ₂₂	42
TS ₂₂	42
P ₂₃	46
R ₂₃	46
TS ₂₃	46
P ₂₄	92
R ₂₄	92
TS ₂₄	92
P ₂₅	88
R ₂₅	88
TS ₂₅	88
P ₂₆	36
R ₂₆	36
TS ₂₆	36
P ₂₇	40
R ₂₇	40
TS ₂₇	40
P ₂₈	60
R ₂₈	60
TS ₂₈	60
P ₂₉	60
R ₂₉	60
TS ₂₉	60
P ₃₀	40
R ₃₀	40
TS ₃₀	40

P ₃₁	40
R ₃₁	40
TS ₃₁	40
P ₃₂	30
R ₃₂	30
TS ₃₂	30
P ₃₃	6
R ₃₃	6
TS ₃₃	6
P ₃₄	34
R ₃₄	34
TS ₃₄	34
P ₃₅	32
R ₃₅	34
TS ₃₅	34
H ₂ CO	4
CH ₄	2
CO	4
PEt ₃	22

R: reactant, P: product, TS: transition state.

Table S4. Total memory requirements (in GB) for local CC single point calculations for the product of reaction 16 of the MOBH35 dataset on two 8-core Intel Xeon E5-2630 v3 CPUs (2.40 GHz).

Methods	Threshold	def2-SV(P)	def2-SVP	def2-TZVP	def2-TZVPP	def2-QZVPP
DLPNO-CCSD(T)	NormalPNO	34.2	44.4	118.8	155.9	195.0
DLPNO-CCSD(T)	TightPNO (TcutPNO= 10^{-6})	41.2	48.4	109.9	136.6	194.0
DLPNO-CCSD(T)	TightPNO (TcutPNO= 10^{-7})	130.4	156.7	194.8	195.9	195.8
DLPNO-CCSD(T)	TightPNO (TcutPNO= 10^{-8})	193.9	195.3	197.3	214.0	252.3
DLPNO-CCSD(T)	veryTightPNO	193.3	193.1	202.2	222.3	252.2
DLPNO-CCSD(T ₁)	NormalPNO	34.5	44.7	118.8	156.6	196.7
DLPNO-CCSD(T ₁)	TightPNO (TcutPNO= 10^{-6})	41.6	48.8	110.6	137.2	202.1
DLPNO-CCSD(T ₁)	TightPNO (TcutPNO= 10^{-7})	130.4	156.9	197.8	197.2	196.6
DLPNO-CCSD(T ₁)	TightPNO (TcutPNO= 10^{-8})	193.9	195.3	197.0	213.9	252.3
DLPNO-CCSD(T ₁)	veryTightPNO	193.3	193.9	201.7	222.9	252.3
PNO-LCCSD(T)	Default	60.0	60.1	60.4	77.3	232.7
PNO-LCCSD(T)	Tight	60.1	60.2	77.3	94.2	233.3
LNO-CCSD(T)	Normal	0.8	1.1	3.6	5.2	10.1
LNO-CCSD(T)	Tight	1.3	2.1	9.6	14.3	35.7
LNO-CCSD(T)	vTight	1.7	3.06	20.8	31.1	99.2
Canonical CCSD(T)		59.7	59.7	173.6	232.0	227.7
Nbasis		184	221	410	506	958

Table S5. Disk usage requirements (in GB) for local CC single point calculations for the product of reaction 16 of the MOBH35 dataset on two 8-core Intel Xeon E5-2630 v3 CPUs (2.40 GHz).

Methods	Threshold	def2-SV(P)	def2-SVP	def2-TZVP	def2-TZVPP	def2-QZVPP
DLPNO-CCSD(T)	NormalPNO	2.9	3.0	12.0	14.0	40.0
DLPNO-CCSD(T)	TightPNO (TcutPNO= 10^{-6})	2.3	1.9	7.5	9.0	33.0
DLPNO-CCSD(T)	TightPNO (TcutPNO= 10^{-7})	6.1	8.3	24.0	30.0	78.0
DLPNO-CCSD(T)	TightPNO (TcutPNO= 10^{-8})	22.0	30.0	139.0	241.0	577.0
DLPNO-CCSD(T)	veryTightPNO	22.0	31.0	141.0	243.0	570.0
DLPNO-CCSD(T ₁)	NormalPNO	15.0	23.0	70.0	78.0	139.0
DLPNO-CCSD(T ₁)	TightPNO (TcutPNO= 10^{-6})	18.0	21.0	50.0	68.0	107.0
DLPNO-CCSD(T ₁)	TightPNO (TcutPNO= 10^{-7})	29.0	46.0	122.0	149.0	287.0
DLPNO-CCSD(T ₁)	TightPNO (TcutPNO= 10^{-8})	57.0	75.0	139.0	242.0	727.0
DLPNO-CCSD(T ₁)	veryTightPNO	57.0	75.0	243.0	324.0	728.0
PNO-LCCSD(T)	Default	10.1	13.4	43.8	54.9	127.7
PNO-LCCSD(T)	Tight	16.5	22.8	81.5	102.6	206.0
LNO-CCSD(T)	Normal	1.1	1.6	3.4	5.3	5.9
LNO-CCSD(T)	Tight	2.5	2.4	5.3	7.2	12.0
LNO-CCSD(T)	vTight	2.6	3.2	8.2	11.0	20.0
Canonical CCSD(T)		12.8	22.5	180.2	325.9	1300.0
Nbasis		184	221	410	506	958

Table S6. Correlation energy recovered from local coupled cluster methods and RMSD values (kcal/mol) for species in MOBH35 with respect to canonical CCSD(T)/def2-SVP.

Methods	Cutoffs	%E _{corr} recovered from CCSD(T)						%E _{corr} recovered from CCSD					
		avg.% ^a	avg.% ^b	avg.% ^c	RMSD ^a	RMSD ^b	RMSD ^c	avg.% ^a	avg.% ^b	avg.% ^c	RMSD ^a	RMSD ^b	RMSD ^c
DLPNO-CCSD(T)	Normal	99.10	99.13	99.14	2.55	1.46	1.39	99.73	99.75	99.76	1.24	0.80	0.71
DLPNO-CCSD(T _i)	Normal	99.42	99.44	99.46	2.38	1.30	1.17	99.73	99.75	99.76	1.24	0.80	0.71
DLPNO-CCSD(T)	Tight	99.41	99.42	99.44	1.95	0.99	0.90	99.71	99.72	99.73	1.11	0.57	0.50
DLPNO-CCSD(T _i)	Tight ^d	99.49	99.51	99.53	2.11	1.04	0.91	99.71	99.72	99.73	1.11	0.57	0.50
DLPNO-CCSD(T _i)	Tight ^e	99.77	99.79	99.80	1.67	0.78	0.56	99.87	99.88	99.89	0.88	0.46	0.34
DLPNO-CCSD(T _i)	Tight ^f	99.90	99.91	99.92	1.20	0.55	0.34	99.96	99.97	99.97	0.57	0.33	0.23
DLPNO-CCSD(T _i)	veryTight	99.89	99.90	99.91	1.14	0.50	0.29	99.94	99.94	99.95	0.55	0.28	0.19
Extrap{6,7}	Tight	99.98	99.99	100.00	1.39	0.70	0.40	99.99	100.00	100.01	0.76	0.45	0.32
Extrap{7,8}	Tight	100.01	100.02	100.03	0.82	0.42	0.27	100.04	100.04	100.04	0.37	0.30	0.26
PNO-LCCSD(T)	Default	99.67	99.68	99.69	1.44	1.11	0.78	99.79	99.80	99.81	0.90	0.76	0.57
PNO-LCCSD(T)	Tight	99.88	99.88	99.88	0.70	0.57	0.28	99.94	99.95	99.95	0.35	0.32	0.13
LNO-CCSD(T)	Normal	99.98	99.99	99.99	0.47	0.32	0.18	99.94	99.94	99.94	0.23	0.23	0.19
LNO-CCSD(T)	Tight	99.99	99.99	100.00	0.26	0.16	0.14	100.00	100.00	100.00	0.12	0.12	0.11
LNO-CCSD(T)	Tight+	100.00	100.00	100.00	0.25	0.16	0.15	100.01	100.01	100.01	0.11	0.11	0.11
LNO-CCSD(T)	vTight	100.00	100.00	100.00	0.14	0.12	0.11	100.02	100.02	100.02	0.08	0.08	0.08

^a All reactions except for 17-20, 24-25; ^b Same but without reaction 9; ^c Same but without reactions 8 and 9; ^d TightPNO with TcutPNO = 10⁻⁶ E_h; ^e TightPNO with TcutPNO = 10⁻⁷ E_h, i.e. the Default TightPNO settings; ^f TightPNO with TcutPNO = 10⁻⁸ E_h.

Table S7. Correlation matrix of the MR character diagnostics.

	T1DIAG	D1DIAG	D2DIAG	FOD_TPSS	FOD_PBE0	%Ecorr[(T)]	Kulik	max{t _{ij} } ^A	max{t _{ij} } ^{AB}	(T1)-(T0)	(T1)/(T0)	1/H-L gap	Matito I _{ND} /I _{tot}	Matito revIND	Truhlar	Mdiag	A100_TPSS	%TAE[ΔX]
T1DIAG	1.0000	0.9081	0.5194	0.4489	0.3422	0.4962	0.8288	0.0381	0.6466	0.5712	0.2756	0.2423	0.1928	0.1922	0.4838	0.2671		
D1DIAG	0.9081	1.0000	0.7063	0.5512	0.5113	0.4816	0.8600	0.1534	0.7596	0.3671	0.4128	0.4328	0.2238	0.2864	0.2897	0.1797		
D2DIAG	0.5194	0.7063	1.0000	0.6386	0.6577	0.5975	0.6208	0.4682	0.5706	0.0298	0.6596	0.5472	0.4294	0.5496	0.1985	0.2659		
FOD_TPSS	0.4489	0.5512	0.6386	1.0000	0.8998	0.4682	0.4882	0.3533	0.4965	0.0848	0.7788	0.4617	0.3052	0.3521	0.0845	0.1155		
FOD_PBE0	0.3422	0.5113	0.6577	0.8998	1.0000	0.4585	0.4473	0.3985	0.6234	-0.1437	0.8454	0.7063	0.3989	0.4636	-0.1339	-0.0185		
%Ecorr[(T)]	0.4962	0.4816	0.5975	0.4682	0.4585	1.0000	0.4936	0.3118	0.5142	0.1786	0.4620	0.5139	0.8587	0.5618	0.5722	0.4654		
max{t _{ij} } ^A	0.8288	0.8600	0.6208	0.4882	0.4473	0.4936	1.0000	0.1677	0.5540	0.4341	0.4576	0.2876	0.2171	0.3146	0.3212	0.2615		
max{t _{ij} } ^{AB}	0.0381	0.1534	0.4682	0.3533	0.3985	0.3118	0.1677	1.0000	0.2300	-0.1212	0.4327	0.3361	0.2827	0.3850	-0.0454	0.0720		
(T1)-(T0)	0.6466	0.7596	0.5706	0.4965	0.6234	0.5142	0.5540	0.2300	1.0000	0.0933	0.3599	0.8392	0.3958	0.5711	0.1054	0.1089		
(T1)/(T0)	0.5712	0.3671	0.0298	0.0848	-0.1437	0.1786	0.4341	-0.1212	0.0933	1.0000	-0.0265	-0.3406	-0.0992	0.0418	0.4945	0.2855		
1/H-L gap	0.2756	0.4128	0.6596	0.7788	0.8454	0.4620	0.4576	0.4327	0.3599	-0.0265	1.0000	0.4470	0.3461	0.4418	-0.0862	-0.0158		
Matito I _{ND} /I _{tot}	0.2423	0.4328	0.5472	0.4617	0.7063	0.5139	0.2876	0.3361	0.8392	-0.3406	0.4470	1.0000	0.5703	0.6799	-0.0596	0.0508		
Matito revIND	0.1928	0.2238	0.4294	0.3052	0.3989	0.8587	0.2171	0.2827	0.3958	-0.0992	0.3461	0.5703	1.0000	0.5144	0.4135	0.2851		
Truhlar	0.1922	0.2864	0.5496	0.3521	0.4636	0.5618	0.3146	0.3850	0.5711	0.0418	0.4418	0.6799	0.5144	1.0000	0.0375	0.1919		
A100_TPSS	0.4838	0.2897	0.1985	0.0845	-0.1339	0.5722	0.3212	-0.0454	0.1054	0.4945	-0.0862	-0.0596	0.4135	0.0375	1.0000	0.8134		
%TAE[ΔX]	0.2671	0.1797	0.2659	0.1155	-0.0185	0.4654	0.2615	0.0720	0.1089	0.2855	-0.0158	0.0508	0.2851	0.1919	0.8134	1.0000		

Table S8. Effects of core-valence correlation on forward and reverse barriers in LNO-CCSD(T) with Tight thresholds (kcal/mol)

def2-TZVPP							cc-pwCVTZ(-PP)							
$V_f^\ddagger, \Delta E_{\text{fwd}}^\#$				$V_r^\ddagger, \Delta E_{\text{rev}}^\#$			$V_f^\ddagger, \Delta E_{\text{fwd}}^\#$			$V_r^\ddagger, \Delta E_{\text{rev}}^\#$			$V_f^\ddagger$	$V_r^\ddagger$
Rxn	no (n-1)sp	with (n-1)sp	$\delta((n-1)\text{sp})$	no (n-1)sp	with (n-1)sp	$\delta((n-1)\text{sp})$	no (n-1)sp	with (n-1)sp	$\delta((n-1)\text{sp})$	no (n-1)sp	with (n-1)sp	$\delta((n-1)\text{sp})$	$\Delta\delta((n-1)\text{sp})^a$	$\Delta\delta((n-1)\text{sp})^a$
1	28.83	26.75	-2.07	15.72	13.90	-1.82	28.54	26.51	-2.03	15.80	14.68	-1.13	0.05	0.69
2	7.28	5.92	-1.36	24.05	22.29	-1.76	7.17	6.17	-1.00	23.89	22.22	-1.67	0.36	0.10
3	1.22	0.98	-0.24	26.19	26.05	-0.14	1.20	1.02	-0.18	25.90	25.83	-0.07	0.06	0.06
4	1.92	1.55	-0.38	7.93	7.76	-0.18	1.96	1.62	-0.34	7.90	7.70	-0.19	0.04	-0.02
5	4.92	4.69	-0.23	22.91	22.64	-0.27	4.71	4.47	-0.23	22.77	22.61	-0.16	0.00	0.10
6	15.46	15.47	0.01	14.71	14.59	-0.12	15.14	15.16	0.02	14.74	14.65	-0.09	0.01	0.03
7	27.57	27.58	0.01	18.88	18.85	-0.02	27.64	27.59	-0.05	18.95	18.96	0.01	-0.06	0.03
8	35.04	35.15	0.12	31.41	31.66	0.26	34.98	34.92	-0.06	31.76	31.81	0.05	-0.17	-0.20
9	27.72	28.32	0.59	12.78	11.59	-1.20	27.76	28.22	0.46	12.84	11.95	-0.89	-0.13	0.31
10	-2.99	-3.96	-0.97	11.05	9.28	-1.77	-3.41	-4.06	-0.65	10.32	9.47	-0.84	0.32	0.93

11	28.80	29.83	1.04	83.16	82.91	-0.25	28.25	28.89	0.64	82.99	82.63	-0.36	-0.40	-0.11
12	5.30	5.47	0.17	37.46	37.39	-0.07	5.32	5.39	0.07	37.49	37.41	-0.08	-0.09	-0.01
13	21.60	20.54	-1.07	49.30	48.58	-0.72	22.32	21.60	-0.72	48.85	48.36	-0.49	0.35	0.23
14	10.36	10.23	-0.14	14.24	14.39	0.15	9.98	9.86	-0.12	14.53	14.63	0.10	0.02	-0.06
15	20.17	20.83	0.65	73.82	74.72	0.89	21.37	21.94	0.57	73.78	74.33	0.55	-0.08	-0.34
16	34.37	35.50	1.12	54.28	53.60	-0.68	35.14	35.94	0.80	53.42	52.85	-0.57	-0.32	0.11
21	8.57	8.17	-0.40	8.57	8.17	-0.40	8.93	8.93	0.00	8.93	8.93	0.00	0.40	0.40
22	14.42	14.43	0.00	27.63	27.32	-0.31	14.34	14.27	-0.08	28.30	28.43	0.13	-0.08	0.44
23	30.22	29.96	-0.26	20.86	20.33	-0.53	30.51	30.13	-0.38	21.02	20.77	-0.25	-0.12	0.28
26	23.91	25.33	1.42	0.10	0.09	-0.01	25.74	26.25	0.50	0.03	0.04	0.01	-0.92	0.02
27	14.28	14.07	-0.22	1.85	1.92	0.08	14.24	14.24	0.00	2.29	2.32	0.03	0.22	-0.05
28	31.07	30.65	-0.42	15.95	15.87	-0.08	31.16	31.26	0.10	15.99	15.92	-0.07	0.52	0.02
29	15.29	15.27	-0.02	31.95	31.97	0.03	15.18	15.22	0.05	32.83	32.99	0.17	0.07	0.14
30	9.92	10.15	0.23	17.78	16.89	-0.88	9.80	9.91	0.11	16.74	16.19	-0.55	-0.12	0.34
31	4.51	3.28	-1.23	12.51	13.38	0.87	3.26	2.72	-0.55	13.36	13.73	0.37	0.69	-0.49
32	20.34	20.50	0.16	62.21	62.00	-0.20	20.53	20.53	-0.01	61.08	60.64	-0.44	-0.16	-0.23
33	1.06	1.20	0.15	9.27	8.11	-1.16	1.06	1.00	-0.07	8.66	8.18	-0.48	-0.22	0.68
34	28.02	29.74	1.72	4.14	3.44	-0.70	29.14	29.93	0.78	3.61	3.33	-0.28	-0.94	0.42
35	16.21	16.74	0.53	-2.16	-2.48	-0.32	16.69	17.13	0.44	-1.81	-1.74	0.08	-0.09	0.40

Table S9. Effects of core-valence correlation on forward and reverse barriers in LNO-CCSD(T) with Tight thresholds (kcal/mol)

def2-{T,Q}ZVPP							cc-pwCV{T,Q}Z(-PP)							
$V_f^\ddagger, \Delta E_{\text{fwd}}^\#$				$V_r^\ddagger, \Delta E_{\text{rev}}^\#$			$V_f^\ddagger, \Delta E_{\text{fwd}}^\#$			$V_r^\ddagger, \Delta E_{\text{rev}}^\#$			$V_f^\ddagger$	$V_r^\ddagger$
Rxn	no (n-1)sp	with (n-1)sp	$\delta((n-1)\text{sp})$	no (n-1)sp	with (n-1)sp	$\delta((n-1)\text{sp})$	no (n-1)sp	with (n-1)sp	$\delta((n-1)\text{sp})$	no (n-1)sp	with (n-1)sp	$\delta((n-1)\text{sp})$	$\Delta\delta((n-1)\text{sp})^a$	$\Delta\delta((n-1)\text{sp})^a$
1	28.69	25.72	-2.97	15.95	14.50	-1.46	28.80	26.41	-2.39	15.93	14.87	-1.06	0.58	0.40
2	7.51	5.71	-1.80	24.59	22.35	-2.24	7.49	6.35	-1.13	24.44	22.45	-1.99	0.67	0.25
3	1.22	0.92	-0.29	26.98	27.60	0.61	1.19	0.99	-0.20	27.00	26.44	-0.56	0.10	-1.17
4	2.17	1.38	-0.79	8.38	8.66	0.27	2.02	1.74	-0.28	8.36	7.87	-0.49	0.52	-0.76
5	5.53	4.90	-0.62	23.18	22.59	-0.58	5.56	5.33	-0.23	22.87	22.70	-0.17	0.39	0.41
6	15.75	15.88	0.12	14.95	14.83	-0.11	15.48	15.50	0.02	14.94	14.84	-0.10	-0.10	0.01
7	27.83	27.80	-0.03	18.78	18.71	-0.07	27.70	27.68	-0.02	18.73	18.61	-0.12	0.01	-0.05
8	34.96	33.51	-1.45	31.99	31.41	-0.58	35.06	N/A	N/A	32.11	32.22	0.11	N/A	0.70
9	29.56	29.56	-0.01	12.95	11.45	-1.50	29.45	29.69	0.25	12.54	11.50	-1.04	0.25	0.47
10	-1.97	-4.59	-2.62	9.23	7.77	-1.47	-2.08	-2.99	-0.91	9.53	8.93	-0.60	1.71	0.87
11	28.06	29.03	0.96	82.26	82.17	-0.09	27.96	28.58	0.62	82.23	82.27	0.04	-0.34	0.13
12	5.36	5.32	-0.04	37.42	37.19	-0.23	5.38	5.48	0.10	37.66	37.77	0.11	0.14	0.34
13	22.01	20.95	-1.06	49.04	48.32	-0.72	21.89	21.44	-0.45	49.45	49.02	-0.43	0.61	0.29
14	10.39	10.18	-0.21	14.70	14.86	0.16	10.16	10.06	-0.10	14.68	14.74	0.06	0.11	-0.10
15	19.62	20.21	0.58	75.44	76.07	0.63	19.97	20.20	0.23	75.55	76.02	0.47	-0.35	-0.16
16	34.58	35.34	0.76	54.79	54.25	-0.55	35.05	35.71	0.66	54.53	54.08	-0.45	-0.10	0.10
21	8.71	8.89	0.18	8.71	8.89	0.18	8.80	9.07	0.26	8.81	9.07	0.26	0.08	0.09
22	14.72	14.15	-0.56	27.81	28.03	0.22	14.60	14.47	-0.12	28.65	29.19	0.54	0.44	0.31
23	30.41	29.84	-0.57	21.16	20.89	-0.27	30.42	29.76	-0.66	21.24	21.10	-0.14	-0.09	0.13
26	24.74	25.54	0.80	0.13	0.17	0.04	26.06	26.31	0.25	0.16	0.14	-0.02	-0.55	-0.06
27	13.78	13.75	-0.02	2.31	2.47	0.16	13.51	13.45	-0.06	2.60	2.63	0.04	-0.04	-0.13
28	30.41	29.85	-0.55	15.69	15.55	-0.13	30.88	31.16	0.28	15.62	15.44	-0.18	0.83	-0.05

29	14.62	14.68	0.06	30.67	30.79	0.12	14.49	14.53	0.04	31.07	31.15	0.08	-0.02	-0.04
30	9.72	9.57	-0.16	17.68	16.97	-0.71	9.72	9.82	0.10	17.20	16.83	-0.38	0.26	0.33
31	4.21	2.85	-1.35	12.31	12.52	0.21	4.37	4.15	-0.23	12.63	12.76	0.13	1.13	-0.08
32	20.21	19.70	-0.51	64.13	64.18	0.05	20.06	20.02	-0.04	63.79	63.51	-0.28	0.47	-0.33
33	0.95	0.83	-0.12	8.95	8.04	-0.92	0.73	0.55	-0.18	8.91	8.72	-0.20	-0.06	0.72
34	27.72	28.27	0.55	3.56	2.84	-0.72	28.35	28.72	0.36	3.59	3.45	-0.14	-0.19	0.58
35	17.49	18.07	0.58	-2.09	-1.75	0.34	17.39	17.74	0.35	-1.96	-1.84	0.12	-0.24	-0.21